



From Fourier to Wavelet

Detecting LLM-Generated Texts Using Wavelet

Group Project II Report

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1 Background

With the continuous development of Large Language Models (LLMs), the texts they generate are becoming increasingly indistinguishable from those written by humans. This distinction is crucial for text source identification, content credibility assessment, and preventing the spread of disinformation. Traditional text detection methods, such as FourierGPT, primarily relies on the Fourier Transform (FFT) to analyze the “uncertainty curve” of texts, comparing the low-frequency components of human and AI texts for classification. However, FFT has significant limitations: it lacks the ability to locate differences in time, leading to information loss, especially when dealing with long texts and

2 Methodology

2.1 Procedure of FourierGPT

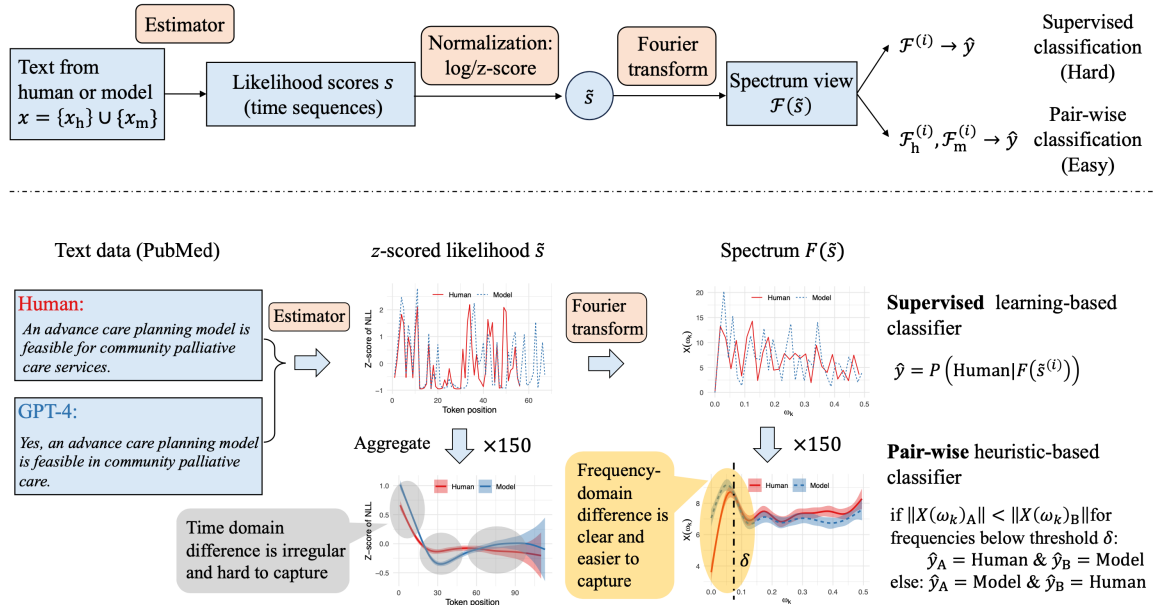


Figure 1: The procedure (above) and example (below) of FourierGPT.

The FourierGPT workflow consists of three main steps: first, it computes negative log-likelihood scores of text using a language model such as GPT; second, these scores are normalized and transformed via Fourier transform to obtain spectral representations;

third, classification is performed using either a supervised learning approach—leveraging mean spectra and SVM or a heuristic method based on comparing frequency component power. This process enables effective discrimination between human-written and model-generated text across different datasets and models.

2.2 Introduction of Wavelet Transform

To address the limitations of FFT, we introduce the Wavelet Transform. Wavelet Transform is a joint time-frequency analysis method that provides both time and frequency information simultaneously. It has the following core advantages:

1. **Time-frequency localization capability:** The parameter b provides precise time location information, while the parameter a offers multi-scale frequency information. The adaptive window allows for wider windows at low frequencies and narrower windows at high frequencies.
2. **Solving FFT issues:** It can separate the contributions of low-frequency energy from different time segments, avoiding information loss due to global averaging and capturing both local details and global structures.

The wavelet transform of a signal $x(t)$ is mathematically defined as:

$$W(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} x(t) \psi^* \left(\frac{t - b}{a} \right) dt$$

where:

- a is the **scale parameter**, inversely related to frequency ($\text{freq} \propto 1/a$),
- b is the **translation parameter**, corresponding to time localization,
- $\psi(t)$ is the **mother wavelet**, a prototype function used to generate a family of wavelets through scaling and translation,
- $\psi^*(t)$ denotes the complex conjugate of the mother wavelet.

2.3 Feature Extraction Strategies

We explored two distinct feature extraction strategies based on the Wavelet Transform, aiming to leverage its time-frequency localization capabilities for capturing meaningful

patterns in textual signals. Each strategy corresponds to a different perspective on how to summarize the wavelet coefficients into features for downstream tasks.

1. Strategy 1: Fix time and compute frequency-domain power sum

This strategy involves selecting a specific time point or segment (i.e., a fixed translation parameter b) and aggregating the energy across all scales (i.e., different values of a).

This approach attempts to characterize the multi-scale frequency behavior at each localized text position. However, it suffers from several drawbacks: poor interpretability, high variance due to local noise sensitivity, and unstable performance in classification tasks. As a result, this strategy was eventually discarded.

2. Strategy 2: Fix scale and compute time-domain power sum

In this strategy, we fix the scale parameter a , effectively focusing on a specific frequency band, and compute the energy distribution across time (i.e., over different values of b). This allows us to assess how a particular frequency pattern evolves across the entire text sequence

This method provides a global view of the signal at a particular scale and enables analysis of semantic consistency across different parts of the text. It exhibits strong interpretability and significantly improved stability and accuracy in downstream detection tasks. Consequently, this strategy was adopted in our final implementation.

2.4 Comparison of DWT and CWT

- **Advantages of DWT:** DWT represents the Discrete Wavelet Transform. It is computationally efficient, controllable feature dimensions, and for approximate scales, high-frequency features can be ignored, and approximate features are global, with the best performance at level=3.
- **Problems of CWT:** CWT represents the Continuous Wavelet Transform. The generated power sequence contains most of the token information from the NLL sequence. For large scales, the global information is too noisy and redundant,

resulting in worse performance than DWT.

2.5 PWH Classification

Building upon the heuristic introduced in the original paper, we adapt the classification method to the wavelet domain, leveraging the time-frequency localization capabilities of the Wavelet Transform. In particular, we exploit the observation that model-generated texts tend to exhibit higher energy in the low-frequency (i.e., high-scale) components of the wavelet spectrum compared to human-written texts.

Wavelet-Based Heuristic. Instead of computing the low-frequency spectrum from the Fourier Transform, we compute the energy across large wavelet scales a_k , which correspond to low-frequency patterns in the text. The classification rule becomes:

$$\sum_{k=1}^{\delta_k} \sum_b |W_{\text{Human}}(a_k, b)|^2 - \sum_{k=1}^{\delta_k} \sum_b |W_{\text{Model}}(a_k, b)|^2 > \varepsilon$$

where:

- a_k denotes the selected large scales corresponding to low frequencies ($\text{freq} \propto 1/a$),
- b is the time translation parameter,
- $W(a_k, b)$ is the wavelet coefficient at scale a_k and time b ,
- δ_k is the number of low-frequency scales included in the comparison,
- ε is a real-valued threshold controlling classification sensitivity.

This rule compares the total low-frequency energy across time between human and model texts. When the human text has significantly greater low-frequency energy, it is classified as such.

Advantages. This wavelet-based variant retains the interpretability of the original heuristic while benefiting from better temporal localization. It is especially effective in identifying global structural differences and stylistic smoothness typically found in model outputs.

3 Experiments

3.1 Test on FourierGPT Dataset

Gen. Model	FFT	Est. Model	Wavelet DWT	δ_k	Wavelet CWT	δ_k
GPT-4	0.9133	GPT2-xl	0.9600	10	0.9267	193
GPT-3.5	0.9467	Mistral	0.9933	12	0.9267	14
GPT-3	0.6867	Mistral	0.7400	92	0.7000	2

Table 1: Detection Results on Pubmed Dataset

Gen. Model	FFT	Est. Model	Wavelet DWT	δ_k	Wavelet CWT	δ_k
GPT-4	0.8467	GPT2-xl	0.9800	8	0.9733	1
GPT-3.5	0.9200	Mistral	1.0000	8	1.0000	1
GPT-3	0.7200	Mistral	0.8867	12	0.9133	3

Table 2: Detection Results on Writing Dataset

Gen. Model	FFT	Est. Model	Wavelet DWT	δ_k	Wavelet CWT	δ_k
GPT-4	0.8733	GPT2-xl	0.8773	10	0.9267	488
GPT-3.5	0.9200	GPT2-xl	0.9933	1	0.9933	2
GPT-3	0.6067	GPT2-xl	0.8400	212	0.9067	189

Table 3: Detection Results on Xsum Dataset

To evaluate the effectiveness of our wavelet-based spectral heuristic, we compare its performance against the original FFT-based method across three benchmark datasets: Pubmed, Writing, and Xsum. As shown in Tables 1–3, both Discrete Wavelet Transform (DWT) and Continuous Wavelet Transform (CWT) consistently outperform the FFT baseline in most settings. Notably, the DWT-based classifier achieves the highest detection accuracy in nearly all cases, with significant improvements observed on **Pubmed** (e.g., +7.8% for GPT-3), **Writing** (+23.2% for GPT-3), and **Xsum** (+38.5% for GPT-3). These results demonstrate the advantage of wavelet methods in capturing localized frequency features, leading to more robust and accurate detection of machine-generated text.

3.2 Test on Ghostbuster Dataset

Beyond the original FourierGPT datasets, we further extended our evaluation to the Ghostbuster benchmark to assess the generalizability of wavelet-based detection. Ghostbuster includes a diverse collection of LLM-generated texts across multiple styles and

topics, making it a more challenging testing ground. Similar to our primary experiments, we applied FFT, DWT, and CWT to extract spectral features for classification.

The results on Ghostbuster reinforce our earlier conclusions: DWT outperforms FFT by a notable margin, especially when the generative model closely mimics human writing. CWT also shows competitive performance in some cases, but suffers from redundancy and instability at high scales. These findings confirm that the superiority of wavelet-based methods is not confined to the original FourierGPT settings, but holds across new, more complex datasets as well.

Dataset	Essay			Reuter			Wp		
Model	FFT	DWT	CWT	FFT	DWT	CWT	FFT	DWT	CWT
Claude	0.611	0.766	0.812	0.809	0.899	0.885	0.619	0.806	0.848
GPT	0.715	0.880	0.823	0.818	0.842	0.793	0.777	0.739	0.644
GPT-prompt1	0.756	0.928	0.888	0.763	0.846	0.841	0.764	0.717	0.707
GPT-prompt2	0.725	0.954	0.911	0.784	0.869	0.910	0.686	0.682	0.772
GPT-semantic	0.711	0.910	0.851	0.763	0.892	0.917	0.707	0.839	0.861
GPT-writing	0.752	0.981	0.960	0.811	0.943	0.937	0.645	0.780	0.714

Table 4: Zero-Shot Classification Accuracy Across Datasets

As shown in Table 4, DWT (Discrete Wavelet Transform) achieves the best accuracy in most cases, particularly on the *Essay* and *Reuter* datasets. This supports the conclusion that wavelet-based transforms capture both local and global text patterns more effectively than FFT.

3.3 Analysis on Value of δ_k

We also analyzed the optimal number of frequency components (δ_k) that lead to peak classification performance. Table 5 shows the corresponding values for each model and dataset. A smaller δ_k suggests that a few dominant frequency components are sufficient for discrimination, often found in datasets with large stylistic differences. Conversely, a larger δ_k indicates the need for broader spectral information, usually for stylistically uniform data.

The variation in δ_k also highlights the time-frequency resolution advantage of the wavelet transform. Specifically, datasets like *Essay*, with distinct human vs. model writing styles, often yield optimal accuracy using only a few low-frequency components. In contrast, more uniform datasets such as *Wp* require broader frequency coverage, especially when

Dataset	Essay			Reuter			Wp		
Model	FFT	DWT	CWT	FFT	DWT	CWT	FFT	DWT	CWT
Claude	18	1	1	280	8	23	25	1	4
GPT	5	15	4	4	16	22	25	186	95
GPT-prompt1	4	13	10	12	15	22	28	208	97
GPT-prompt2	4	16	4	5	13	23	38	187	21
GPT-semantic	5	14	7	13	9	26	11	41	2
GPT-writing	5	2	6	5	10	25	344	87	2

Table 5: Optimal δ_k : Number of Frequency Components for Best Accuracy

using FFT. This further validates the effectiveness of DWT in capturing meaningful discriminative patterns.

Summary on Experiments. Through experimental data comparison, our method demonstrated excellent performance on both the original paper dataset and the new batch of data (including reuter, wp, and essay parts). The introduction of Wavelet Transform effectively solved the time localization problem of FFT. The feature extraction strategy of fixing frequency and calculating time power sum significantly improved detection accuracy. In the comparison of DWT and CWT, DWT showed better performance in computational efficiency and feature dimension control, with the best decomposition effect at level=3, where the approximate coefficient A3 (freq=0) had the highest distinguishability.

4 Future Work Directions

4.1 Algorithm Optimization

1. Explore custom wavelet bases suitable for NLP tasks. The current wavelet transform may not fully utilize local time information, and further exploration can be conducted on the three-dimensional time-frequency graph.
2. Research more efficient wavelet transform algorithms to further improve detection accuracy and efficiency.

4.2 Application Expansion

1. Extend this method to multi-language text detection to verify its applicability and effectiveness in different language environments.

2. Explore the application of wavelet transform in other NLP tasks, such as text generation quality assessment and machine translation quality detection.

5 Summary

This project's LLM-generated text detection method based on Wavelet Transform has effectively improved detection accuracy and stability by addressing the limitations of FFT. In the future, we will continue to optimize the algorithm and expand its applications to provide more powerful tools for text detection in the field of natural language processing.

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